

Cube Values

Teacher's Edition

Standard and Targets

MA.8.AR.2.3 Given an equation in the form of $x^2 = p$ and $x^3 = q$, where p is a whole number and q is an integer, determine the real solutions.

Boundary Cube roots of perfect cubes from -125 to 125 . Students are not required to isolate the x^3 term.

Learning targets I can cube a number. I can evaluate a cube root, including of a negative. I can solve $x^3 = q$ and say why there is exactly one solution.

Essential questions What does it mean to cube a number? Why does a cube root of a negative number exist when a square root of one does not? Why does $x^3 = q$ have one solution when $x^2 = p$ has two?

Building on Lessons 1 and 2 — squaring, perfect squares, radicals, and solving $x^2 = p$.

Working toward MA.8.NSO.1.2, plotting and comparing square roots and cube roots, and MA.8.NSO.1.7, order of operations with radicals.

Where this sits in the course Third and last of three consecutive lessons on MA.8.AR.2.3. This is also the lesson with the sharpest evidence attached to it: on the 2025 released items, **“select all solutions to $x^3 = 64$ ” was answered correctly by 26% of students** — one of the six lowest scores on the whole Grade 8 FAST. Slides 13–15, worksheet question 11 and the whiteboard sequence all aim at that item directly.

Materials Whiteboards and markers. Calculators are allowed but every value in the required questions is on the short list of perfect cubes. Worksheet 7A.02.03. No cubes to build — the isometric drawings on the slides and the sheet do the work.

Lesson at a Glance

Windy Hill periods run **53 minutes**.

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up: same and different	2	4 min
Notes I–IV: cubing, the perfect cubes and their negatives, negative cube roots, exactly one solution	3–6	13 min
Example 1: two cube roots	7–9	5 min
Example 2: solving $x^3 = q$	10–12	5 min
Example 3: same 64 — one solution or two?	13–15	5 min
Whiteboards: 9 questions, each with its reveal	16–33	15 min
IXL — start today's skills	34	5 min
Total		53 min

Where the lesson lives: Slides 13–15. It is the only place in the unit built around a published score, and the score is 26%.

Read this before you plan the day. Math Nation's component times total 45 to 50 minutes before any worksheet, and the lesson is built as two partner Explorations plus a Guided Instruction. The plan above fits 53 because both Explorations become Notes I and II with Examples 1 and 2 as the worked versions, and their Guided Instruction table becomes Notes III.

No partner or group work anywhere in this lesson. Math Nation's version has five printed instances plus a Think-Pair-Share named in the teacher tips. All five are now teacher-led questions or independent whiteboard rounds.

Vocabulary

cube (a number) — to multiply the number by itself three times

perfect cube — a number you get by cubing an integer

cube root — a number that gives the radicand when multiplied by itself three times

volume — the number of unit cubes that fill a solid

index — the small number in the notch — it is 3 for a cube root, and it is left off altogether for a square root

A note on the glossary. Math Nation lists no key terms and no additional terms for this lesson, then introduces “cube root” in the middle of it. The five above are the ones the worksheet uses. “Index” reappears from Lesson 2 because this is the first time students see one written down.

Slide-by-Slide Teaching Notes

Slide 1. Cube Values *1 min*

- Post the learning target and give the contrast straight away.
- Yesterday the number of solutions changed with the sign of p .
- Today it never changes — it is always one — and only the SIGN of the answer moves.
- That contrast is the lesson, and Example 3 is where it gets settled.

Slide 2. Warm-Up *4 min*

- TYPE: Same and different.
- INDEPENDENT.
- ANSWERS: same — both start from 64, both ask for one length, both answers are whole numbers. Different — a square has two dimensions and a cube has three. Question 3: the same total splits into two equal factors in one case and three in the other, so the factors have to be bigger in the two-factor case. Take two or three answers before confirming; the whole lesson is smuggled into question 3. This replaces Math Nation's magic-square puzzle, which had arithmetic in it but nothing to do with cubes.
- OFF THE SLIDE, YOURS TO SAY: Same 64. Different question. That is today in one line.

Slide 3. Notes: I. Cubing means multiplying a number by itself three times 3 min

- Students copy all four lines.
- Item C is Lesson 1's misconception with a 3 in it, and students who fixed it for squares often un-fix it for cubes.
- Write $4^3 = 12$ on the board, ask what is wrong, correct it.
- Item D is what makes the whole lesson concrete: the volume of a cube is the thing being asked for every time.
- OFF THE SLIDE, YOURS TO SAY: Same rule as Lesson 1, one factor further along.

Slide 4. Notes: II. The perfect cubes, and their negatives 3 min

- Read both rows aloud.
- The negative row is the one that matters and it has no counterpart on the square side — you cannot square an integer and get -8 .
- Point at the panel and say you will explain it in two minutes; it is the hinge between the two lessons.
- OFF THE SLIDE, YOURS TO SAY: There is no negative list for perfect squares. That is Notes III.

Slide 5. Notes: III. A negative number has a cube root 4 min

- THE NEW IDEA IN THIS LESSON.
- Item D is the whole explanation and it is arithmetic, not a rule: write the three factors out in full and cross the first two negatives off against each other, leaving one.
- Then contrast with yesterday — with two factors there is nothing left over, so the answer cannot be negative.
- The panel is the sentence to say twice.
- OFF THE SLIDE, YOURS TO SAY: So a cube root of a negative number exists. A square root of one does not.

Slide 6. Notes: IV. Solving $x^3 = q$ — always exactly one 3 min

- Students copy the table.
- Put yesterday's three-case table beside this one on the board if it is still up.
- Yesterday the COUNT changed with the sign of p ; today the count never changes and only the SIGN of the answer moves.
- The error to watch for is a student writing ± 2 for $x^3 = 8$, carrying yesterday's habit forward — have them cube -2 and watch it fail.
- OFF THE SLIDE, YOURS TO SAY: Yesterday $x^2 = p$ gave two, one or none. Today $x^3 = q$ always gives one.

Slide 7. Example 1 1 of the 5 min

- SPOKEN ONLY.
- No pencils.
- The right-hand one, because the radicand is negative and the index is odd.
- Getting that said before any arithmetic is the point — Notes IV says the sign of the answer follows the sign of what is inside, and this is the first chance to use it as a prediction rather than a check.

Slide 8. Example 1 2 of the 5 min

- Work both on the board. 125 and -64 are both inside the benchmark's range, and so is every value on today's worksheet apart from two marked STRETCH at the end.
- The panel is worth saying aloud: yesterday every equation answer carried a plus-or-minus, and today none of them do.
- OFF THE SLIDE, YOURS TO SAY: Both answers are single numbers. There is no $\pm$ anywhere on this slide.

Slide 9. Example 1 2 of the 5 min

- Do this at the board, physically crossing off the first two minus signs.
- The leftover one is the entire explanation and it generalizes: any odd number of negative factors gives a negative product, any even number gives a positive.
- That single fact decides every sign question left in this unit.
- OFF THE SLIDE, YOURS TO SAY: Count the negative factors. An odd count leaves one behind, and that is the whole rule.

Slide 10. Example 2 1 of the 5 min

- Whiteboards, nothing revealed.
- Expect some boards to say ± 3 — that is yesterday's habit and it is exactly what the next two slides take apart.
- Do not correct it verbally; let the arithmetic do it.

Slide 11. Example 2 2 of the 5 min

- Substitute in front of them.
- This is the half students accept easily.
- The next slide is the half they do not.

Slide 12. Example 2 2 of the 5 min

- THE SLIDE THAT DOES THE WORK.
- A student who wants to write ± 3 needs to see 3 fail, not be told it fails.
- Cube it out loud and land on 27.
- Do this once here and both the whiteboard round and the worksheet's select-all get easier.
- Worksheet question 9, $y^3 = -125$, is this same item; question 8 is its mirror, with a positive right-hand side and -2 the value to reject.
- OFF THE SLIDE, YOURS TO SAY: Substituting the other sign and watching it FAIL is what proves there is only one answer. Being told is not the same thing.

Slide 13. Example 3 1 of the 5 min

- On the 2025 released items, a select-all question asking for the solutions of $x^3 = 64$ was answered correctly by 26% of students — one of the six lowest scores on the entire Grade 8 FAST, and it is this benchmark.
- Take the vote and record it on the board; you will come back to it.

Slide 14. Example 3 2 of the 5 min

- Build it one row at a time and let the third row do the work: the same move that succeeded on the left fails on the right, and the reason is the count of negative factors from Example 1.

- Do not state the conclusion — the next slide is the conclusion and they should reach it first.

Slide 15. Example 3 *2 of the 5 min*

- Say plainly that the test will ask this and will offer -4 as a choice.
- Go back to the vote you recorded and ask who has changed their mind and why.
- Worksheet question 11 is the released item's format and question 13 asks them to write the explanation in sentences.
- OFF THE SLIDE, YOURS TO SAY: Two negative factors make a positive. Three make a negative. That single fact decides every answer in this unit.

Slide 16. Whiteboards • Question 1 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 17. Whiteboards • Question 1 of 9 — reveal *about 1 minute with the discussion*

- The baseline.
- Everything after this changes exactly one thing about it.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name.

Slide 18. Whiteboards • Question 2 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 19. Whiteboards • Question 2 of 9 — reveal *about 1 minute with the discussion*

- One feature changed: the sign of the radicand.
- Odd count of negative factors, so the answer is negative.
- Notes III.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name.

Slide 20. Whiteboards • Question 3 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 21. Whiteboards • Question 3 of 9 — reveal *about 1 minute with the discussion*

- One feature changed from question 1: the index is 2 instead of 3.
- Same 64, different answer, and it is bigger — a smaller number repeated three times reaches 64, so two copies have to be larger.
- That is the warm-up.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name.

Slide 22. Whiteboards · Question 4 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 23. Whiteboards · Question 4 of 9 — reveal *about 1 minute with the discussion*

- THE ONE THAT EARNS ITS PLACE.
- One feature changed from question 3: the sign of the radicand — the same change that took question 1 to question 2, where it worked fine.
- Here it does not, and the difference is the index.
- Two negative factors cancel completely; three leave one behind.
- A board saying -8 has applied question 2's move under a square root.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name.

Slide 24. Whiteboards · Question 5 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 25. Whiteboards · Question 5 of 9 — reveal *about 1 minute with the discussion*

- One feature changed: an equation instead of the symbol.
- Still one answer.
- This is the released item that 26% of Florida's students got right, and the choice they take is -4 .
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name.

Slide 26. Whiteboards · Question 6 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 27. Whiteboards · Question 6 of 9 — reveal *about 1 minute with the discussion*

- One feature changed from question 5: the exponent.
- Now there are two.
- Close by putting questions 5 and 6 side by side and asking which one word in the question changed the number of answers — the exponent, and nothing else on the line.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name.

Slide 28. Whiteboards · Question 7 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 29. Whiteboards · Question 7 of 9 — reveal *about 1 minute with the discussion*

- One feature changed: a real object, so the answer carries a unit.
- The volume is in CUBIC inches and the edge is in inches — Notes II, and it is the same distinction lesson 2.01 made between square units and units.
- THE BOARD TO LOOK FOR SAYS 5 CUBIC INCHES, which is the right number wearing the wrong unit, and a FAST item will offer it.
- The other board to watch divides 125 by 3 and answers 41.67; ask that student to multiply their answer by itself three times and see what they get.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name.

Slide 30. Whiteboards · Question 8 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 31. Whiteboards · Question 8 of 9 — reveal *about 1 minute with the discussion*

- One feature changed from question 5: the right-hand side is negative.
- STILL ONE ANSWER, and this is the pairing that matters — $x^2 = -27$ would have none at all, and $x^3 = -27$ has exactly one.
- Three negative factors leave a negative behind; two cancel.
- THE BOARD TO LOOK FOR SAYS ± 3 , which is question 6's habit arriving under a cube, and the check kills it in one line: $3 \times 3 \times 3$ is 27, not -27 , so 3 is not a solution and never was.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name.

Slide 32. Whiteboards · Question 9 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 33. Whiteboards · Question 9 of 9 — reveal *about 2 minutes with the closing discussion*

- WRITTEN WORK, and the only question in the round that asks for it.
- With the worksheet gone this is the only place in the period the habit gets built, and IXL takes an answer without ever seeing the work.
- ANSWER: no. $\sqrt[3]{-64} = -4$, because $(-4)(-4)(-4) = -64$.
- HER RULE IS RIGHT FOR SQUARE ROOTS AND SHE HAS APPLIED IT ONE INDEX TOO FAR, which is exactly what questions 3 and 4 of this round were built to separate — and the question asks for that, not just for "no". $\sqrt{-64}$ really does have no real value; $\sqrt[3]{-64}$ is -4 .
- THE REASON IS COUNTABLE, so make them write it: two negative factors cancel and leave a positive, three leave one behind.
- A student who writes "no, it is -4 " has the answer and has not said when Priya would have been right, which is the half of this that will be on the test.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name.

Slide 34. IXL 5 min

- IXL.
- The skills are assigned in Google Classroom, not named here, and they are deliberately not ours -- Croix: "The students need questions that aren't exclusively mine."
- Makes them think." Circulate.
- The thing to watch for is a student answering in the box with no paper in front of them: IXL takes an answer and never sees the work, so this is the one part of the period where the habit can quietly lapse.

Worksheet 7A.02.03 — Answer Key

1. the table	1, 8, 27, 64, 125 and -1, -8, -27, -64, -125 The bottom two rows are the point. A negative side length is not a real box, but $(-s)^3$ is a real number and it is negative. Squares never do this.
2. the cube	64 cubic units $4^3 = 4 \times 4 \times 4$. A student who answers 12 has multiplied by 3; one who answers 16 stopped after squaring.
3. volume 125	5 inches 125 is the largest perfect cube the FAST will ask about, and -125 the smallest.
4. $\sqrt[3]{27}$	3
5. $\sqrt[3]{-64}$	-4 Unlike a square root, a cube root of a negative exists. Three negative factors give a negative product.
6. $\sqrt[3]{125}$	5
7. $\sqrt[3]{-1}$	-1
8. $x^3 = 8$	$x = 2$ $(-2)^3 = -8$, not 8, so -2 is not a solution. This is the single most-missed idea on this benchmark.
9. $y^3 = -125$	$y = -5$
10. $n^3 = 0$	$n = 0$
11. select all	4 and $\sqrt[3]{64}$ Two of six. This exact question was on the 2025 released items and 26% of students got it right. The trap is -4 and $-\sqrt[3]{64}$, both of which give -64. 8 is $\sqrt{64}$; $21\frac{1}{3}$ is $64 \div 3$.
12. the table and graph	$(-2, -8)$ $(-1, -1)$ $(0, 0)$ $(1, 1)$ $(2, 8)$ All five sit on gridlines. Compare with $y = x^2$ from Lesson 1: that graph stayed on or above the axis, this one crosses it.
13. why one, not two	Two negative factors make a positive; three make a negative. The point needs the count of negative factors, not just the two answers. "Because cubes are different" explains nothing.
14. $\sqrt[3]{216}$ (stretch)	6 Outside the tested range. Same method.
15. $m^3 = 343$ (stretch)	$m = 7$ Still one solution.

Question 11 is the first of the two to circulate for; question 13 is the other, below. It is the released item. Two boxes are correct and one of them is a radical — students who tick 4 and stop have not noticed that $\sqrt[3]{64}$ is 4. Students who also tick -4 have carried Lesson 2's habit across, which is the whole point of slides 13–15.

Question 13 is the other, and it is the one to read tonight. The point requires the count of negative factors, not the two answers. It is the difference between a student who can do this benchmark and one who has memorized two facts about 64.

Questions 14 and 15 are marked STRETCH on the student sheet. They use 216 and 343, which sit outside MA.8.AR.2.3's stated range of -125 to 125 . They are there because an accelerated class can handle the arithmetic, and they are labeled so that nobody — student or substitute — mistakes them for required fluency.

Additional Practice 7A.02.03 — Answer Key

1. the table	1, 8, 27, 64, 125 and -1, -8, -27, -64, -125 Cubing preserves the sign. Squaring does not.
2. the cube	27 cubic meters A student who answers 9 stopped after squaring.
3. volume 8	2 centimeters
4. $\sqrt[3]{1}$	1
5. $\sqrt[3]{-27}$	-3 A cube root of a negative exists. A square root of one does not.
6. $\sqrt[3]{64}$	4
7. $\sqrt[3]{-125}$	-5
8. $x^3 = 27$	$x = 3$ $(-3)^3 = -27$, so -3 does not work.
9. $a^3 = -8$	$a = -2$
10. $w^3 = 1$	$w = 1$ Not ± 1 . $(-1)^3 = -1$.
11. select all	5 and $\sqrt[3]{125}$ Two of six. -5 and $-\sqrt[3]{125}$ both give -125. 25 is 5^2 , and $4\frac{1}{3}$ is $125 \div 3$.
12. the table and graph	$(-2, -8)$ $(-1, -1)$ $(0, 0)$ $(1, 1)$ $(2, 8)$ The graph crosses the x-axis at the origin; $y = x^2$ touches it there and turns back up.
13. why one, not two	Two negative factors make a positive; three make a negative. The point needs the count of negative factors.
14. $\sqrt[3]{343}$ (stretch)	7
15. $m^3 = 512$ (stretch)	$m = 8$

This sheet is optional. Post it to Google Classroom for students who want more. It is not assigned and it is not collected. Its stretch pair uses 343 and 512.

Changes from the Math Nation Lesson

- **The cube values were brought inside the benchmark's range.** MA.8.AR.2.3 states its boundary explicitly: perfect cubes from -125 to 125 . Math Nation's homework and practice sheets use 216, 343, 512 and 1,728 — six of their twenty cube items sit outside the range the test will use, plus one in the guided table. Every required question on these sheets is now inside the range, and two larger cubes are kept at the end of each sheet under a STRETCH heading that says so in print.
- **A false premise was corrected on both student sheets.** Math Nation's question 4 tells students “the graph of $y = x^2$ only has positive y-values” and asks them to explain why. At $x = 0$ the value is 0, which is not positive. The comparison question is now worded around what the two graphs do at the origin, and the same correction was made in Lesson 1.
- **A grid that could not hold its own answer was rebuilt.** Math Nation's coordinate plane for $y = x^3$ runs from $+9$ down to -7 , but the table asks students to plot $(-2, -8)$. In their printed key the red dot for that point sits in white space below the last gridline. The grid on these materials runs -9 to 9 vertically and -3 to 3 horizontally, which is the range the table actually needs.
- Every partner and group activity is replaced — five printed instances plus the Think-Pair-Share named in the teacher tips.
- The warm-up was replaced. Math Nation's is a 3×3 magic square: a fine puzzle with no connection to cubes, keyed “answers will vary” when there are exactly eight arrangements. The new warm-up states two facts — a square with area 64 has a side of 8, a cube with volume 64 has an edge of 4 — and asks what is the same, what is different, and why the same 64 gives two different answers — which is slides 13–15 in miniature.
- **The Wrap-Up was kept, with one line removed.** Math Nation's Ticket Out the Door for this lesson is the only one in the three lessons that produces real gradeable mathematics, so it stays in substance. The instruction that students “should not use a calculator” was dropped, because it cuts against your standing policy and nothing on the ticket needs one anyway.
- A select-all item was added to the worksheet and a select-all round to the whiteboards, because that is the format of the released item this lesson is judged on.
- Answer keys were built for both sheets from the same source file as the student versions, so the two cannot drift apart.
- **Nothing else in the mathematics was changed.** The retuned cube values are the one place a number differs from a question that was KEPT, and the reason is printed on the student page. The items listed above as added or replaced — the select-all, the warm-up — carry their own numbers, and none of their answers was taken from the published sheet.